

Data Analyst Questions & Dashboard Objectives

Python Data Analyst Questions

- 1 Find the top 10 brands generating the highest revenue using `final_price` and visualize the result using Matplotlib or Seaborn.
- 2 Create a customer return analysis model to identify which category and location have the highest return percentage.
- 3 Perform time-series analysis on `purchase_date` to identify monthly sales trends and seasonal spikes.
- 4 Build a correlation heatmap between price, discount, rating, `review_count`, `seller_rating`, and `shipping_time_days`.
- 5 Find products where `rating > 4.5`, `discount > 30%`, and `review_count > 100`, then analyze which category performs best.
- 6 Create a seller performance score using `seller_rating`, return rate, delivery delay, and average product rating.
- 7 Detect outliers in product pricing using IQR or Z-score method and identify suspicious products.
- 8 Build a predictive model to predict whether a product will be returned using classification algorithms.
- 9 Analyze how `shipping_time_days` affects customer ratings and delivery status.
- 10 Create an advanced customer purchasing behavior analysis based on device, `payment_method`, location, and category purchased.

SQL Data Analyst Questions

- 1 Find the top 5 categories with the highest total sales revenue.
- 2 Calculate the return percentage for each brand and rank them from highest to lowest.
- 3 Find sellers whose average seller rating is above 4.5 but have delayed deliveries above average.
- 4 Write a query to find monthly sales growth percentage using window functions.
- 5 Find the top-selling subcategory in each category using `ROW_NUMBER()`.
- 6 Identify products where `stock < 50`, `review_count > average review_count`, and `rating > 4`.
- 7 Find cities with the highest average shipping time and compare their average customer ratings.
- 8 Use CTEs to calculate total revenue, total returns, and average discount category-wise.
- 9 Find which payment method contributes the highest revenue and lowest return rate.
- 10 Write a query to classify sellers into Excellent, Good, Average, and Poor based on `seller_rating` and delivery performance.

Power BI Data Analyst Questions

- 1 Create a dynamic sales dashboard showing revenue, profit estimation, return percentage, and average rating.
- 2 Build DAX measures for Total Revenue, Average Discount, Return Rate, and Average Shipping Time.
- 3 Create a drill-through report from category to subcategory to brand.
- 4 Build a customer behavior dashboard analyzing device usage, payment methods, and locations.
- 5 Create a KPI dashboard to monitor seller performance.

- 6 Build a decomposition tree to analyze reasons behind high product returns.
- 7 Create a forecasting visual for monthly sales trends.
- 8 Analyze delayed deliveries by category and seller using conditional formatting.
- 9 Create a report comparing actual price, discounted price, and rating impact.
- 10 Create a report that identifies low-stock high-demand products using DAX and visual indicators.

Excel Data Analyst Questions

- 1 Create a Pivot Table dashboard showing category-wise revenue and returns.
- 2 Use SUMIFS, COUNTIFS, and AVERAGEIFS to analyze seller performance.
- 3 Create an interactive dashboard using slicers for category, location, payment method, and device.
- 4 Use conditional formatting to highlight delayed deliveries, low-rated products, and high return products.
- 5 Create a dynamic monthly sales trend chart using Pivot Charts.
- 6 Use advanced formulas like INDEX-MATCH, XLOOKUP, and IFERROR to generate automated reports.
- 7 Build a customer return analysis report using Pivot Tables and charts.
- 8 Create a stock management tracker showing products likely to go out of stock.
- 9 Analyze the relationship between discounts and ratings using scatter charts.
- 10 Create an executive summary dashboard with KPIs including Total Sales, Average Rating, Return Rate, and Average Delivery Time.

Independent Dashboard Questions

- 1 What KPIs should be included in an e-commerce sales dashboard for management-level reporting?
- 2 How can you design a dashboard to identify high-return products and their root causes?
- 3 What visualizations should be used to compare sales across categories, brands, and locations?
- 4 How can a dashboard help identify delayed delivery patterns across cities and sellers?
- 5 What metrics should be included to monitor seller performance effectively?
- 6 How can customer behavior be visualized using device type, payment method, and location?
- 7 What should be included in a stock monitoring dashboard to avoid inventory shortages?
- 8 How can a dashboard help management identify profitable vs non-profitable product categories?
- 9 Which dashboard elements can help detect customer satisfaction issues using ratings and reviews?
- 10 How can forecasting visuals and trend analysis improve business decision-making in the dashboard?

What Can Be Achieved From This Dataset

- 1 Identify top-performing categories and brands.
- 2 Analyze customer purchasing behavior.
- 3 Improve seller performance monitoring.
- 4 Reduce product return rates.
- 5 Optimize inventory and stock management.

- 6 Improve delivery efficiency and logistics.
- 7 Understand the impact of discounts on sales.
- 8 Detect low-performing products and sellers.
- 9 Predict future sales trends and customer demand.
- 10 Create business intelligence dashboards for strategic decision-making.